MENTAL MATH

MULTIPLYING BY 11

2-DIGIT & 3-DIGIT

2-DIGIT: 68 × 11 = 748

STEP 1: 6 [___] 8

STEP 2:

① carry

6 4 ← 6 + 8 = 14

STEP 3: 7 4 8

= 748

3-DIGIT: 689 × 11 = 7,579

STEP 1: 6 [___] [___] 9

STEP 2:

① ① carry

8 + 9 = 17

6 4 7 9

6 + 8 = 14

STEP 3: 7 5 7 9

= 7,579

FUTURISTIC ACADEMY LEARNING

STEP 2: COMPUTING THE SECOND PARTIAL PRODUCT (86 × 8)

BREAKING DOWN THE MULTIPLICATION

86 × 8

Break 86 into 80 and 6

FIRST PART: MULTIPLY TENS

80 × 8

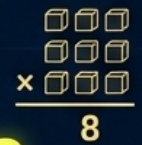


8 × 8 = 64

80 × 8 = 640

SECOND PART: MULTIPLY ONES

6 × 8



6 × 8 = 48

COMBINING THE RESULTS

640 + 48 = 688

TOTAL PARTIAL PRODUCT

= 688

FUTURISIC ACADEMY

MENTAL MATH MULTIPLICATION

STEP 3: COMBINING THE PARTIAL PRODUCTS

5160

+

688

Combine Results

Partial Product 1

5160

+ 688

5848

Calculate Total Sum

FINAL ANSWER

Mental Strategy

CHECKED

VEDIC CROSS MULTIPLICATION: 86 x 68

$$\begin{array}{r} 86 \\ \times\ 68 \\ \hline \end{array}$$

STEP 1: UNITS (6x8)

Tens	Units
	6
	8

6 x 8 = 48

Carry 4 → 8 (result)

Result so far: __ 8

STEP 2: CROSS MULTIPLICATION

8 — 8
6 — 6

(8x8) + (6x6) + 4 (Carry)

(8 x 8 = 64) (6 x 6 = 36)

64 + 36 + 4 = 104

Result: 4 (middle digit)

Carry 10

Result so far: _ 48

STEP 3: TENS (8x6)

Tens	Units
8	
6	

(8 x 6) + 10 (Carry)

48 + 10 = 58

58 (leftmost digits)

FINAL RESULT: 5848